

INT 10: BIOS Video services

AH=0: set video mode (clear screen) AL=0: text 40*25 16 color (mono) 1: text 40*25 16 color 2: text 80*25 16 color (mono) 3: text 80*25 16 color 4: CGA 320*200 4 color 5: CGA 320*200 4 color (m) 6: CGA 640*200 2 color 7: MDA monochrome text 80*25 8: PCjr 9: PCjr 0A: PCjr 0B: reserved 0C: reserved 0D: EGA 320*200 16 color 0E: EGA 640*200 16 color 0F: EGA 640*350 mono 10: EGA 640*350 16 color 11: VGA 640*480 16 color 12: VGA 640*480 16 color 13: VGA 320*200 256 color NB: add 80h to mode to prevent screen clr	
AH=1: set cursor shape CH,CL= starting,ending scan line = 6, 7: default CGA cursor =11,12: default EGA cursor =13,14: default VGA cursor CH =32(20h): no cursor	
AH=2: set cursor location BH= display page (normal=0) DH,DL= row,col (0..24,0..39/79)	40:50 is updated
AH=3: report cursor location, shape BH= display page (normal=0)	CH,CL=starting,ending scan line DH,DL=row,column
AH=4: light pen service	
AH=5: select active page AL= page 0..	
AH=6: scroll up (clear screen) AL= lines (=0: clear screen) BH= attribute (=7: white on black) CH= upper line (=0: top) CL= left col (=0: far left) DH= lower line (=18(24): bottom) DL= right col (=4F(79): far right) DX=184F(6223)	
AH=7: scroll down	
AH=8: read 'char'&attrib at cursor BH=display page (0)	text & graphic! modes AH,AL = attr,char
AH=9: write 'char'&attrib AL=char BH=display page (0) BL=attribute CX=no. of repetitions =2000(7D0h) to fill 80*25 screen	cursor not moved
AH=A: write 'char' AL=char BH=display page (0) =background pixel attrib (mode 13h) BL=foreground pixel attrib (graphics) CX=no. of repetitions	

AH=B: set palette BH,BL=0,color (0-15): text: screen border color CGA : border & background color (if EGA/VGA: use INT 10,10) BH,BL=1,palette (0-1): CGA 320*200: palette ,0: green,red,brown ,1: cyan,magenta,white	40:66 is updated
AH=C: write pixel AL=color 0..3 - bit 7 set: XOR CX=column 0..319/639 DX=row 0..199	
AH=D: read pixel CX=column 0..319/639 DX=row 0..199	AL=pixel color 0..3
AH=E: write 'char' teletype AL=char (7,8,10,13: special action) BH=page(0) BL=foreground color (graphic)	moves cursor
AH=F: read video mode	AH=no.of display text columns AL=video mode =01:text 40*25 color =03:text 80*25 color (standard DOS) =04:CGA 320*200 4-color =06:CGA 640*200 2-color =10:EGA 640*350 16 color =12:VGA 640*480 16 color BH=current video page (0)

AH=10: EGA/VGA BIOS	
AL= 0: Set Palette Register BL= Palette register 0..15 BH= Palette register value 0..63 : rgbRGB = 0 00 000000 black = 1 01 000001 blue = 2 02 000010 green = 3 03 000011 cyan = 4 04 000100 red = 5 05 000101 magenta =14 20 010100 brown = 7 07 000111 white =56 38 111000 dark gray =57 39 111001 light blue =58 3A 111010 light green =59 3B 111011 light cyan =60 3C 111100 light red =61 3D 111101 light magenta =62 3E 111110 yellow =63 3F 111111 bright white AL= 1: Set Border Register BH= border color 0..63 AL= 2: Set Palette and Border Registers ES:DX->17 byte table of values 0-15: values (0..63) for registers 16: value (0..63) for border AL= 3: Select Blink or Background BL= 0: no blink, 16 background colors = 1: blink, 8 background colors AL= 7: Read Palette Register BL= palette register no. 0..15 AL= 8: Read Border Register	BH=palette register value 0..63 BH=border color 0..63

AL=20: Load 8*8 Graphic Character Set ES:BP->table of bit patterns (hi-bit only)	
AL=21: Load Custom Graphic Character Set CX= bytes/character BL= 0: DL=no. of rows = 1: 14 rows = 2: 25 rows = 3: 43 rows ES:BP->table of bit patterns	
AL=22: Load 8*14 Graphic Character Set BL= 0: DL=no. of rows = 1: 14 rows = 2: 25 rows = 3: 43 rows	
AL=23: Load 8*8 Graphic Character Set BL= 0: DL=no. of rows = 1: 14 rows = 2: 25 rows = 3: 43 rows	
AL=24: Load 8*16 Graphic Character Set BL= 0: DL=no. of rows = 1: 14 rows = 2: 25 rows = 3: 43 rows	
AL=30: Get Character Set Information BH= 0: 8*8 CGA hi-bit characters (128-255) = INT 1F ptr (in BIOS: C000:460A) = 1: Current graphics characters = INT 43 ptr (in BIOS: C000:420A) or (in HMA : FFFF:BF32) = 2: 8*14 characters (in HMA) = 3: 8*8 characters (in HMA) = 4: 8*8 hi-bit characters (in BIOS) = 5: 9*14 characters = 6: 8*16 characters = 7: 9*16 characters	ES:BP->table of bit patterns CX = bytes/character (8,14,16) DL = character rows on screen

AH=12: ALTERNATE SELECT	
BL=10: return EGA information	BH=0,1: color,mono BL=0,1,2,3: 64,128,192,256k EGA RAM CH=EGA feature connector input status CL=EGA switch settings
BL=20: Alternate Print Screen	- for rows > 25
BL=30: Select Text Scan Line No. AL= 0: 200 lines (CGA emulation) = 1: 350 lines (EGA emulation) = 2: 400 lines (VGA default)	
BL=31: Palette Loading on Mode Change AL= 0: enable default palette loading = 1: disable palette loading - keep current values	
BL=32: Buffer/Port Addressing AL= 0: enable addressing = 1: disable addressing	- used with BL=35: Switch Display
BL=33: Grey Scale AL= 0: enable grey scale = 1: disable grey scale	
BL=34: Cursor Emulation AL= 0: enable CGA-type cursor = 1: disable cursor	

BL=35: Switch Display AL= 0: initial adaptor off = 1: initial adaptor on = 2: turn active video off = 3: turn active video on ES:DX->128 byte buffer	- used with BL=32: Buffer/Port Addressing enables switching between PS/2 video adaptors
BL=36: Refresh AL= 0: enable refresh = 1: disable refresh (faster screen)	

AH=13: Write text string AL= 0: without moving cursor = 1: moving cursor = 2: without moving cursor - c/a string = 3: moving cursor - c/a string BH= page (0) BL= attrib (AL=0,1) CX= string length DH,DL= row,column for cursor start ES:BP->string buffer	- c/a string: string of c/a words char,attrib,char,attrib,... - c/a word: hi: attrib lo: char
AH=1A: Video Combination Code AL= 0: Get	BL= active display code BH= inactive display code = 0: no display 1: MDA (monochrome) 2: CGA 4: EGA color 5: EGA mono 6: PGC 7: VGA mono 8: VGA color B: MCGA mono C: MCGA color FF: unknown
AL= 1: Set BL= active display code BH= inactive display code - codes as for Get above	
AH=1B: Get BIOS Functionality Info BX= 0 ES:DI->64 byte buffer	video mode and configuration table at ES:DI
AH=1C: VGA State AL= 0: get buffer size CX= state = 0,1,2: DAC,BIOS,registers	BX= no. of 64-byte buffer blocks required
AL= 1: store CX= state (as above) ES:BX->buffer	- use after AL=0: get buffer size
AL= 2: restore CX= state (as above) ES:BX->buffer	- use after AL=1: store

VESA calls: Somerson, DOS5PT.729

AH=4F: VESA	
AL= 0: Return Super VGA Information ES:DI->256 byte buffer	AX=004F if VESA-compatible ES:DI data at offset: 00: "VESA" 04: version number 06: address of vendor ID 0E: address of mode list 12: amount of memory on VGA

AL= 1: Return Mode Information CX= Super VGA mode ES:DI->256 byte buffer	ES:DI data at offset: 00: mode attributes 02: window A attrib (readable/writeable) 03: window B attrib 04: window granularity (offset of each bank in Kb) 06: window size (bank size in granularity units) 08: address of window A 0A: address of window B 0C: address of window function 10: length of one scan line in bytes - optional info -: 12: horizontal resolution 14: vertical resolution 16: character width in pixels 17: character height in pixels 18: number of bit planes 19: bits per pixel (includes planes) 1A: banks (interleaved memory segments) 1B: memory model 1C: bank size (memory segment size) 1D: number of available pages 1E: reserved
AL= 2: Set Super VGA Mode	
AL= 3: Return Super VGA Mode	
AL= 4: Save/Restore Super VGA State	
AL= 5: Bank Switching	
AL= 6: Set/Get Logical Scan Line Length	
AL= 7: Set/Get Start of Display Memory	